

RATIO

- Used for comparing two or more similar quantities.

- Notation- a:b, where a= antecedent, b=consequent

Value of ratio won't change, when we multiply or divide it by a constant number.

Value of ratio will change, when we add or subtract a constant number.

- Commonly used terms:

i) Duplicate and Triplicate Ratio

The duplicate ratio of a:b is $(a^2:b^2)$ and the triplicate ratio of a:b is $(a^3:b^3)$.

ii) Sub-duplicate and Sub-Triplicate Ratio

The Sub-duplicate ratio of a:b is $\sqrt{a}:\sqrt{b}$ and the triplicate ratio of a:b is $\sqrt[3]{a}:\sqrt[3]{b}$

PROPORTION

Four quantities a, b, c, d are said to be in proportion if

$a:b = c:d$ or $a:b :: c:d$.

Continued proportion:

$$a:b = b:c = c:d, \quad \frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

Mean Proportion:

If $a:b = b:c$

Then b is called mean proportional, if $b^2 = ac$ or $b = \sqrt{ac}$

Third Proportion:

If $a:b = b:c$

Then c is called Third proportional, if $c = \frac{b^2}{a}$

Fourth Proportion:

If $a:b = c:d$

Then, d is fourth proportional to numbers a, b, c if, $d = \frac{bc}{a}$

EXERCISE

1. The ratio of two numbers is 11:19. If the first number is 132 then, find the second number.
2. Two numbers are in the ratio 14:17. If sum of these two numbers is 248 then, find the value of both the numbers.
3. The ratio of two numbers is 13:21. If difference of the two numbers is 576 then, find the value of both the numbers.
4. A and B have a total of Rs.850 with them. They decided to distribute this amount in the ratio 8:9. Find the amount received by A.
5. Two persons X and Y have Rs.1350 with them. They decided to distribute this amount in the ratio 12:13. Find the amount received by Y.
6. If $x:y = 2:3$ and $y:z = 3:4$ then, find $x:y:z$.
7. If $x:y = 3:4$ and $y:z = 9:10$ then, find $x:y:z$.
8. If $x:y = 7:8$ and $y:z = 5:3$ then, find $x:y:z$.
9. If $a:b = 3:2$, $b:c = 5:3$, $c:d = 4:7$ then, find $a:b:c:d$.
10. If $a:b = 1:4$, $b:c = 2:3$, $c:d = 5:6$ then, find $a:b:c:d$.
11. If $(a + b) : (b + c) : (c + a) = 8 : 7 : 5$ and $(a + b + c) = 50$. Find the value of c .
12. If $p : q = r : s = t : u = 2 : 3$ then, find $(mp + nr + ot) : (mq + ns + ou)$.
13. If $a/(b+c) = 2/3$ and $b/(a+c) = 3/4$ then, find $c/(a+b)$.
14. If $(b+c)/a = 7/6$ and $(a+c)/b = 6/5$ then, find $(a+b)/c$.
15. If $(x - y + z) : (y - z + 2w) : (2x + z - w) = 2 : 3 : 5$ then, $(3x + 3z - 2w) : w = ?$
16. Find the Mean Proportion of 4 & 16.
17. Find the Mean Proportion of 7 & 28.
18. Find the Third Proportion of 18 & 24.
19. Find the Third Proportion of 25 & 45.
20. Find the Fourth Proportion of 9, 36 and 12.
21. Find the Fourth Proportion of 7, 11 and 35.
22. The ratio of two numbers X and Y is 9:10. If 18 is added to both of them the ratio becomes 11:12. Find the original value of X.
(a) 81 (b) 72
(c) 54 (d) 90
23. The students in three classes are in the ratio 4:6:9. If 15 students are increased in each class then, the ratio changes to 7:9:12. Find the total number of students in the three classes before the increase?
(a) 95 (b) 76
(c) 100 (d) 114
24. Two numbers are in the ratio 3:5. If 6 is added to both of them, the ratio becomes 2:3. The numbers are?
(a) 21 and 35 (b) 30 and 50
(c) 24 and 40 (d) 18 and 30
25. What number should be subtracted from both the terms of the ratio 11:15 so as to make it as 2:3?
(a) 2 (b) 3
(c) 4 (d) 5
26. The income of A and B is in the ratio 5:3 while the ratio of their expenditure is 3:1. If at the end of the year each saves Rs.800 then, find the income of A.
(a) 4000 (b) 2000
(c) 3000 (d) 2800
27. The income of A and B is in the ratio 3:2 and the ratio of their expenditure is 4:3. If their savings are Rs.2000 and Rs.1000 then, what is the income of B?
(a) 6000 (b) 5000
(c) 4000 (d) 3000

28. A planet is divided into two parts Land and Water and the ratio of Land and Water on that planet is 3:2 while the ratio of Land and Water in its Northern Hemisphere is 2:3. What is the ratio of Land and Water in the Southern Hemisphere?

- (a) 3:4 (b) 4:1
(c) 3:2 (d) 5:2

29. A planet is divided into two parts Land and Water and the ratio of Land and Water on that planet is 2:3 while the ratio of land and water in its Northern Hemisphere is 3:4. Find the ratio of Land and Water in the Southern Hemisphere.

- (a) 13:22 (b) 11:17
(c) 12:19 (d) 14:23

30. A bag contains Rs.2240 in the form of Rs.5, Rs.2 and Rs.1. The coins are in the ratio 3:4:5. Find the no. of Rs.5 coins.

- (a) 300 (b) 240
(c) 320 (d) 400

31. A bag contains 250 coins of the denomination Rs.1, 50p and 25p. The ratio of their amount is 10:6:7. Find the number of Rs.1 coins.

- (a) 60 (b) 50
(c) 80 (d) 100

32. A person broke a Pencil into two parts such that the ratio of length of larger part to the smaller part is same as the ratio of length of the whole pencil to the larger part. Find the ratio of larger part to the smaller part.

- (a) $\frac{\sqrt{5}+1}{2}$ (b) $\frac{\sqrt{3}+3}{5}$
(c) $\frac{\sqrt{5}+3}{4}$ (d) $\frac{\sqrt{7}+3}{2}$

33. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$ then, which of the following statements is correct?

- (a) Each fraction is equal to $\frac{1}{2}$ only.
(b) Each fraction is equal to 1 or -1
(c) Each fraction is equal to 1 or $\frac{1}{2}$
(d) Each fraction is equal to $\frac{1}{2}$ or -1

PARTNERSHIP

34. A, B and C are three partners and they invested Rs.2000, Rs.5000 and Rs.7000 in a business. At the end of the year, they earned a profit of Rs.42000. Find the share of each.

- (a) 6000,15000,21000
- (b) 8000,10000,21000
- (c) 6000,8000,15000
- (d) 9000,12000,18000

35. X and Y jointly started a business. X invested 4 times that of Y. Also, X invested his money for double the time of Y. If Y earned Rs.100 as his share then, the total profit earned was?

- (a) 800 (b) 900
- (c) 1000 (d) 1100

36. Three partners X, Y and Z invested an amount in the ratio 5:7:8. If X invested for 14 months, Y for 8 months and Z for 6 months then, find the ratio of their profit.

- (a) 35:28:24 (b) 28:24:35
- (c) 24:35:28 (d) 35:24:28

37. Virat and Dhoni invested a sum in the ratio of 5:6. After 6 months, Rohit joined the business with an amount equal to Dhoni's. Find the ratio of their profits after the end of the year.

- (a) 3:5:6 (b) 5:3:6
- (c) 6:5:3 (d) 5:6:3

38. P: Q: R invested in the ratio 8:5:7. At the end of the year, they share their profits in the ratio 4:1:5. Find the ratio of time for which they invested.

- (a) 50:14:35 (b) 14:50:35
- (c) 35:14:50 (d) 35:50:14

PROBLEMS BASED ON AGES

40. Ratio of ages of 'A' and 'B' is 9:10. After 4 years it will be 11:12. Find the present age of 'A'.

- (a) 12 (b) 15
- (c) 18 (d) 21

41. Ratio of age of A and B is 9:16. After 10 years it will be 2:3. Find the sum of their ages.

- (a) 40 (b) 60
- (c) 50 (d) 70

42. 10 years ago the ratio of ages of A and B was 3:1. After 10 years it will be 5:3. Find their present age.

- (a) 40 & 20 (b) 40 & 30
- (c) 30 & 10 (d) 50 & 30

43. 8 years ago A's age was 4 times the age of his son's age. After 8 years A's age will be twice of his son's age. Find their present age.

- (a) 80 and 16 (b) 80 and 32
- (c) 40 and 16 (d) 40 and 32

44. Difference of ages of A and B is 16 years. 6 years ago, A's age was thrice of B's age. Find their present age.

- (a) 30 and 14 (b) 24 and 8
- (c) 32 and 16 (d) 40 and 24

45. 'X' says to his son that when you were born, I was of your current age then, what was the age of the son 5 years ago if the age of 'X' is 36 years?

- (a) 18 (b) 13
- (c) 11 (d) 10